

Project Documentation

AI-Powered Bug Detection using Python

Objective

Develop a basic Python application that automatically analyzes Python source code and detects syntax-related bugs using Python's built-in AST module.

Project Scope

The application scans a Python file, identifies syntax errors, displays the error details, and generates a bug report for users.

Tools & Technologies

Python 3.x, AST (Abstract Syntax Tree), Visual Studio Code, Git, GitHub.

Project Structure

sourcecode/bug_detector.py
sourcecode/sample_code.py
README.md
requirements.txt
output/bug_report.txt
screenshots/
documentation/

Working Principle

1. Read the Python source file.
2. Parse the code using the AST module.
3. Detect syntax errors.
4. Display the error information.
5. Save the bug report to bug_report.txt.

Features

Automatic syntax checking, bug reporting, simple user interface, and file-based output.

Expected Output

The application reports whether syntax errors exist. If an error is found, it displays the line number, error type, and description, and stores the report in a text file.

Future Enhancements

Add machine learning models for smarter bug detection, support multiple programming languages, detect logical errors, and provide automatic fix suggestions.

Conclusion

This project demonstrates a basic AI-inspired bug detection system that automates syntax analysis and generates a clear bug report. It serves as a foundation for more advanced static code analysis tools.